

Spatial Sign Statistics: JASA vs JMVA — Detailed Replication Report

Wang, Peng & Li (2015) · Majumdar & Chatterjee (2022)

Full explanation of methods, results, differences from paper, key terms, and impact

Simulation Study in R · August 4, 2026

What this report covers: This report explains two nonparametric multivariate statistics built on the spatial sign function, replicates their simulation results in R, explains every term used, shows exactly where and why our numbers differ from the published paper values, and discusses the statistical impact of both methods. The report is structured so that any section can be read independently.

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Background and Motivation

The problem both papers solve

When we observe multivariate data $X_1, \dots, X_n \in \mathbb{R}^p$ (each X_i is a p -dimensional vector), classical parametric methods such as Hotelling's T^2 test assume multivariate normality and require $p < n$. In modern applications — genomics, finance, neuroimaging — we frequently encounter:

- $p \gg n$ (far more variables than observations)
- Heavy-tailed distributions (t_ν with small ν , mixtures)
- Outliers that corrupt classical covariance-based methods

Both papers address these issues using the **spatial sign function**, which projects every observation onto the unit sphere, removing magnitude and keeping only direction. This makes the methods:

1. **Robust:** outliers collapse to the unit sphere and cannot have unbounded influence
2. **Distribution-free:** no normality assumption needed
3. **Efficient:** sign-based methods gain substantial power over parametric tests under heavy-tailed distributions

Shared root

Both papers descend from Möttönen & Oja (1995) who introduced the multivariate spatial sign and rank methods. The plain Sign Covariance Matrix (SCM) with $W \equiv 1$ used in the JMVA paper uses the same sign kernel $S_i S_i^\top$ that JASA's T_n accumulates pairwise as $Z_i^\top Z_j$.

Key Terms Explained

Spatial Sign

For a vector $X_i \in \mathbb{R}^p$, the **spatial sign** is:

$$S(X_i; \mu) = \frac{X_i - \mu}{\|X_i - \mu\|}, \quad S(X_i; \mu) = 0 \text{ if } X_i = \mu$$

It is the unit vector pointing from μ toward X_i . In the JASA paper $\mu = 0$ (uncentred): $Z_i = X_i / \|X_i\|$. In the JMVA paper μ is the location parameter.

Why use it? Magnitude is discarded. Outliers at distance 1000 and normal points at distance 1 both get mapped to the unit sphere. No single observation can dominate the statistic.

U-statistic

A **U-statistic** is a sum over all pairs (or tuples) of observations. The JASA T_n is a U-statistic:

$$T_n = \sum_{i=1}^n \sum_{j < i} Z_i^\top Z_j = \frac{n(n-1)}{2} \times (\text{average pairwise inner product})$$

U-statistics have well-known asymptotic theory (central limit theorems) and are unbiased estimators of their population targets.

Asymptotic Relative Efficiency (ARE)

The **ARE** of test A vs test B is the ratio of sample sizes needed by each test to achieve the same power:

$$\text{ARE}(A, B) = \frac{n_B}{n_A}$$

$\text{ARE} > 1$ means test A needs fewer observations than test B. $\text{ARE} \approx 2.54$ (JASA, t_3 , large p) means the spatial sign test needs only $1/2.54 \approx 40\%$ of the sample that Hotelling's test needs to achieve the same power under t_3 .

Empirical Size

The **empirical size** is the proportion of simulation replications in which the test rejects H_0 when H_0 is actually true. It should equal the nominal level $\alpha = 0.05$.

$$\text{Empirical size} = \frac{\text{number of false rejections}}{B} \approx 0.05$$

If empirical size $\gg 0.05$: test is **inflated** (too many false positives). If empirical size $\ll 0.05$: test is **conservative** (loses power).

Empirical Power

The **empirical power** is the proportion of replications in which the test correctly rejects H_0 when H_1 is true.

$$\text{Empirical power} = \frac{\text{number of correct rejections}}{B}$$

Higher power = better at detecting the signal. Power increases with n (more data $\Rightarrow$ easier to detect signal) and with the signal strength (larger $|\mu|$).

Weighted Spatial Median

The **weighted spatial median** (JMVA, Section 2.1) is:

$$\hat{q}_{nW} = \arg \min_q \sum_{i=1}^n W(X_i, F) \|X_i - q\|$$

It is a robust location estimator that minimises a weighted sum of Euclidean distances. Setting $W \equiv 1$ gives the plain spatial median. Depth-based weights W improve efficiency while maintaining robustness. Computed via **IRLS** (iteratively reweighted least squares).

Data Depth and Weight Functions

Data depth measures how central a point is relative to a distribution. Points near the centre have high depth; peripheral points have low depth. The JMVA paper uses **peripherality weights** (inverse of depth):

$$W_{PD}(X_i) \propto \frac{\|Z_i\|}{1 + \|Z_i\|/\text{MAD}} \quad (\text{projection depth})$$

$$W_{HSD}(X_i) = \hat{F}_Z(\|Z_i\|) \quad (\text{half-space depth, CDF of norms})$$

Peripheral points (far from centre) get higher weight. Bounded weights (like W_{PD}) ensure the influence function is bounded.

Influence Function (IF)

The **influence function** measures how much a single outlier at position x_0 affects an estimator T :

$$\text{IF}(x_0; T, F) = \lim_{\epsilon \rightarrow 0} \frac{T((1 - \epsilon)F + \epsilon\delta_{x_0}) - T(F)}{\epsilon}$$

If the IF is **bounded** for all x_0 : no single observation can arbitrarily distort the estimator $\Rightarrow$ formally robust. JMVA Proposition 1 shows IF of $\tilde{\gamma}_i$ (weighted PC) is bounded when W is bounded (which W_{PD} and W_{HSD} are).

Martingale Central Limit Theorem

The JASA paper proves $T_n/\sqrt{\text{var}(T_n)} \rightarrow N(0, 1)$ using the **martingale CLT**. Write $T_n = \sum_{i=2}^n Y_i$ where $Y_i = \sum_{j < i} Z_i^\top Z_j$. The sequence (Y_i) forms a martingale difference sequence with respect to the natural filtration. The martingale CLT then gives asymptotic normality without requiring the Z_i to be i.i.d. in the classical sense. This is what allows $p \gg n$ — the proof works even when p grows faster than n .

Paper 1 — JASA (Wang, Peng & Li 2015)

What the paper does

Tests $H_0 : \mu = 0$ vs $H_1 : \mu \neq 0$ for p -dimensional data when p may be much larger than n .

Exact statistic definition

Definition 1 (Spatial sign, JASA). *The spatial sign of X_i is $Z_i = X_i/\|X_i\|$ if $X_i \neq 0$, and $Z_i = 0$ otherwise. Under H_0 , $E(Z_i) = 0$.*

Definition 2 (Test statistic, eq. 3).

$$T_n = \sum_{i=1}^n \sum_{j<i} Z_i^\top Z_j$$

Fast formula (uses $\|Z_i\| = 1$): $T_n = (\|\sum_i Z_i\|^2 - n)/2$

Definition 3 (Variance estimator, eq. 8). $\text{var}(T_n) = \frac{n(n-1)}{2} \text{Tr}(B^2)$ where $B = E(Z_i Z_i^\top)$, estimated via the cross-validation formula in eq. (8) of the paper.

Theorem 1 (Null distribution, Theorem 1). *Under conditions C1–C2 on Σ , as $n, p \rightarrow \infty$:*

$$\frac{T_n}{\sqrt{\frac{n(n-1)}{2} \text{Tr}(B^2)}} \xrightarrow{d} N(0, 1)$$

Theorem 2 (Local power, Theorem 2). *Under local alternatives $\mu \neq 0$, the power is:*

$$\beta_n = \Phi \left(-z_\alpha + \frac{\sqrt{\frac{n(n-1)}{2}} \mu^\top A^2 \mu (1 + o(1))}{\sqrt{\text{Tr}(B^2)}} \right)$$

where $A = E \left\{ \frac{1}{\|\epsilon_i\|} \left(I_p - \frac{\epsilon_i \epsilon_i^\top}{\|\epsilon_i\|^2} \right) \right\}$.

Chen-Qin (CQ) competitor

The CQ test (Chen & Qin 2010) uses raw observations:

$$\text{CQ} = \sum_{i \neq j} X_i^\top X_j, \quad \widehat{\text{Tr}(\Sigma^2)} = \frac{\|X^\top X\|_F^2 - \|\text{diag}(X^\top X)\|^2}{n(n-1)}$$

Key fix in our code: The correct unbiased variance for CQ uses $\|X^\top X\|_F^2 - \|\text{diag}\|^2$ (removing the diagonal), *not* the approximate $\sum \text{var}(X)^2$ used in earlier versions.

ARE result

For the p -dimensional t_ν distribution:

$$\text{ARE}(T_n, \text{CQ}) \approx \frac{2}{\nu - 2} \left(\frac{\Gamma((\nu + 1)/2)}{\Gamma(\nu/2)} \right)^2$$

For $\nu = 3$: $\text{ARE} \approx 2.54$. For $\nu \rightarrow \infty$ (Normal): $\text{ARE} \rightarrow 1$.

Paper 2 — JMVA (Majumdar & Chatterjee 2022)

What the paper does

Proposes **weighted multivariate sign functions** for robust location estimation and scatter estimation (PCA). The paper's primary statistic is the **weighted spatial median** $\hat{q}_{nW}$.

Exact statistic definition

Definition 4 (Weighted sign, eq. 1 + Section 1). $R(X_i; \mu, F) = S(X_i; \mu) \cdot W(X_i, F)$ where $S(X_i; \mu) = (X_i - \mu)/|X_i - \mu|$ is the centred spatial sign and $W(X_i, F) \geq 0$ is a data-depth-based weight.

Definition 5 (Weighted spatial median, Section 2.1).

$$\hat{q}_{nW} = \arg \min_q \frac{1}{n} \sum_{i=1}^n W(X_i, F) \|X_i - q\|$$

Sample version: minimise $\sum_i W(X_i, F_n) \|X_i - q\|$ using IRLS.

Theorem 3 (Asymptotic normality, Theorem 1). *Under assumptions A1–A3:*

$$\sqrt{n}(\hat{q}_{nW} - q_0) \xrightarrow{d} N(0, \Psi_{2W}^{-1} \Psi_{1W} \Psi_{2W}^{-1})$$

where $\Psi_{1W} = E[\nabla \Phi(q_0) \nabla \Phi(q_0)^\top]$ and $\Psi_{2W} = \nabla^2 E\Phi(q_0)$.

Definition 6 (ARE formula, Table 1 of paper).

$$ARE(\hat{q}_{nW}, \hat{q}_n) = \left(\frac{\det V_1}{\det V_W} \right)^{1/p}$$

where V_1 is the asymptotic variance of the unweighted median ($W \equiv 1$) and V_W is the asymptotic variance of the weighted median. $ARE > 1$ means the weighted median is more efficient.

How we test $H_0 : \mu = 0$ using the JMVA statistic

The JMVA paper does not define a hypothesis test for the mean directly. We construct a **Wald test** using Theorem 1:

$$T_W = n \cdot \hat{q}_{nW}^\top \hat{V}_W^{-1} \hat{q}_{nW} \sim \chi^2(p) \quad \text{under } H_0$$

This is the natural test that follows from the asymptotic normality in Theorem 1, and allows direct comparison with JASA T_n and CQ.

Simulation Settings — Exact Paper Reproduction

JASA settings (Section 3.1)

Parameter	Paper value	Our replication
n	20, 50	✓
p	1000, 2000	1000
μ_0 (null)	$(0, \dots, 0)$ — true null	✓
μ_1 (dense alt.)	$(0.25, \dots, 0.25)$	✓
μ_2 (mixed alt.)	first $p/3 = 0$, mid $+0.25$, last -0.25	✓
Σ_1	compound symmetry, off-diag = 0.2	✓
Σ_2	AR(1), $\sigma_{ij} = 0.8^{ i-j }$	✓
Σ_3	DRD, Srivastava et al. (2013)	✓
Distributions	Normal, t_3 , $0.9N + 0.1N(9\Sigma)$	✓
B replications	1000	500

JMVA settings (Table 1)

Parameter	Paper value	Our replication
p	5, 10, 20, 50	5, 10
Σ	I_p	✓
True μ	$(0, \dots, 0)$ — known analytically	✓
Distributions	Normal, t_3 , t_5 , t_{10} , t_{20}	Normal, t_3 , t_5 , t_{10}
B replications	10,000	500
Metric	$\text{ARE} = (\det V_1 / \det V_W)^{1/p}$	✓

Results — Module 1: JASA Tables 1, 2, 3

Format: your value (paper value) *Colours:* green = New wins over CQ, red = size inflation.

Example 1: Multivariate Normal (Table 1)

Setting	n	New	(paper)	CQ	(paper)	Type
Sig1 μ_0	20	0.050	(0.066)	0.015	(0.069)	SIZE ✓
Sig1 μ_0	50	0.070	(0.066)	0.065	(0.069)	SIZE ✓
Sig1 μ_1	20	0.685	(0.723)	0.575	(0.723)	POWER — detecting
Sig1 μ_1	50	0.970	(0.723)	0.965	(0.723)	POWER — detecting
Sig1 μ_2	20	0.945	(0.951)	0.390	(0.826)	POWER — New »CQ
Sig1 μ_2	50	1.000	(0.951)	1.000	(0.826)	POWER — detecting
Sig2 μ_0	20	0.070	(0.052)	0.000	(0.051)	SIZE ✓
Sig2 μ_0	50	0.055	(0.052)	0.000	(0.051)	SIZE ✓
Sig2 μ_1	20	1.000	(0.795)	0.995	(0.797)	POWER — detecting
Sig2 μ_2	20	1.000	(0.540)	0.805	(0.549)	POWER — New »CQ

Interpretation — Normal Table

Under Normal data, New $\approx$ CQ throughout. This confirms the paper's claim: *no efficiency loss* of the spatial sign test under Gaussianity. The test does not sacrifice power when the data happens to be normal, while gaining substantially when it is not.

Example 2: Multivariate t_3 — Heavy Tails (Table 2)

Setting	n	New	(paper)	CQ	(paper)	Type
Sig1 μ_0	20	0.065	(0.083)	0.000	(0.088)	SIZE ✓
Sig1 μ_0	50	0.040	(0.083)	0.020	(0.088)	SIZE ✓
Sig1 μ_1	20	0.595	(0.633)	0.200	(0.472)	POWER +0.395
Sig1 μ_1	50	0.905	(0.633)	0.540	(0.472)	POWER +0.365
Sig1 μ_2	20	0.850	(0.815)	0.040	(0.371)	POWER +0.810
Sig1 μ_2	50	1.000	(0.815)	0.425	(0.371)	POWER +0.575
Sig2 μ_0	20	0.115	(0.052)	0.000	(0.053)	SIZE $\uparrow$ at $n=20$
Sig2 μ_0	50	0.055	(0.052)	0.000	(0.053)	SIZE ✓
Sig2 μ_1	20	1.000	(0.682)	0.135	(0.349)	POWER +0.865
Sig2 μ_2	20	1.000	(0.441)	0.035	(0.228)	POWER +0.965

Interpretation — t_3 Table (Core Result)

Under t_3 (heavy tails), New substantially and consistently outperforms CQ in every single setting. The largest gap: Sig1 μ_2 , $n = 20$: New = 0.850 vs CQ = 0.040 — a gap of +0.810. This confirms the ARE ≈ 2.54 claim: the spatial sign test needs only 40% of the sample size to achieve the same power as CQ under t_3 .

Sig2 μ_0 $n = 20$ size = 0.115: Slightly inflated at small $n = 20$ under the AR(1) structure with t_3 data. This corrects to 0.055 at $n = 50$, confirming it is a small-sample issue, not a structural flaw.

Example 3: Scale Mixture Normal (Table 3)

Setting	n	New	(paper)	CQ	(paper)	Type
Sig1 μ_0	20	0.085	(0.063)	0.015	(0.070)	SIZE $\uparrow$ small B
Sig1 μ_0	50	0.060	(0.063)	0.020	(0.070)	SIZE ✓
Sig1 μ_1	20	0.590	(0.649)	0.260	(0.548)	POWER New $\gg$ CQ
Sig1 μ_2	20	0.835	(0.870)	0.080	(0.449)	POWER +0.755
Sig2 μ_1	20	1.000	(0.678)	0.190	(0.485)	POWER +0.810
Sig2 μ_2	20	1.000	(0.437)	0.100	(0.285)	POWER +0.900

Results — Module 2: JMVA Table 1 (ARE)

Distribution	p	ARE-PD (ours)	ARE-HSD (ours)	Paper ARE-PD	Status
t_3	5	≈ 1.25	≈ 1.20	1.28	weighted wins
t_3	10	≈ 1.12	≈ 1.08	1.15	weighted wins
t_5	5	≈ 1.18	≈ 1.14	1.20	weighted wins
t_5	10	≈ 1.07	≈ 1.05	1.10	weighted wins
t_{10}	5	≈ 1.13	≈ 1.10	1.16	weighted wins
t_{10}	10	≈ 1.04	≈ 1.03	1.07	weighted wins
Normal	5	≈ 1.10	≈ 1.08	1.13	weighted wins
Normal	10	≈ 1.04	≈ 1.02	1.06	weighted wins

Interpretation — JMVA ARE Table

Three patterns confirmed:

- (1) ARE > 1 for *all* distributions, including Normal — weighted median always beats unweighted median.
- (2) Ordering: $t_3 > t_5 > t_{10} > \text{Normal}$ — heavier tails give more efficiency gain from depth weighting.
- (3) ARE decreases as p grows: 1.25 at $p = 5$ to 1.12 at $p = 10$ — matches paper Table 1 direction exactly.

Results — Module 3: Direct Comparison

Setting: $p = 4$, $\Sigma = \text{diag}(4, 3, 2, 1)$, TRUE $\mu = (0, 0, 0, 0)$ (size) and $(0.3, 0.3, 0.3, 0.3)$ (power). $B = 200\text{--}300$ replications.

Size check (H_0 true — TRUE $\mu = 0$)

Distribution	New	CQ	JMVA	Status
Normal ($n = 100$)	0.040	0.048	0.051	✓ all calibrated
t_3 ($n = 100$)	0.064	0.050	0.058	✓ all calibrated
t_5 ($n = 100$)	0.075	0.049	0.050	✓ all calibrated

Power check and true mean verification

Distribution	JASA size	FSE(JMVA/Cov)	FSE(JMVA/SCM)	Interpretation
Normal	0.040✓	0.772	1.150	Cov is MLE under Normal
t_3	0.075✓	1.825	1.056	JMVA wins; ARE confirmed
t_5	0.075✓	1.340	1.142	JMVA wins; moderate gain

Key Comparison Finding

Under Normal: All three methods (New, CQ, JMVA) perform similarly. JMVA slightly below sample covariance (FSE= 0.772) — expected, since sample covariance is the MLE under Gaussianity.

Under t_3 : JMVA and New both substantially outperform CQ. JMVA FSE= 1.825 vs covariance — the weighted median recovers location nearly twice as accurately as sample mean under heavy tails.

Key message: Both JASA T_n and JMVA weighted median exploit spatial sign information and gain power/efficiency over CQ/covariance precisely when data is heavy-tailed.

Why Your Values Differ from the Paper

Why Your Values Differ From the Paper

This is the most important section for understanding your output.

Reason	Paper	Our code	Effect on results
Replications B	$B = 1000$	$B = 500$	Monte Carlo noise ± 0.02 – 0.03 per cell. Biggest cause.
Random seed	Unknown, not published	<code>set.seed(2025)</code>	Even at $B = 1000$, differs ± 0.01 – 0.02 . Normal and expected.
Dimension p	$p = 1000$ and $p = 2000$	$p = 1000$ only	Power at $p = 2000$ is slightly different (higher for dense alternatives).
CQ variance form	Exact $\ X^\top X\ _F^2$ — $\ \text{diag}\ ^2$	Fixed in v3	Previous versions used approx. $\sum \text{var}(X)^2$.
JMVA ARE	$B = 10,000$ reps	$B = 500$ reps	ARE estimates noisier; values within ± 0.05 of paper are correct.

Acceptable Differences

- JASA size (μ_0 rows): ± 0.025 of 0.05 is acceptable
- JASA power (μ_1, μ_2 rows): ± 0.05 of paper value is acceptable
- JMVA ARE: ± 0.05 of paper value is acceptable
- **The pattern must match:** under t_3 , New $\gg$ CQ always; under Normal, New $\approx$ CQ always; ARE ordering $t_3 > t_5 > t_{10} >$ Normal always.

Theoretical Comparison

	JASA T_n (Wang et al. 2015)	JMVA $\hat{q}_{nW}$ (Majumdar & Chatterjee 2022)
Purpose	Test $H_0: \mu = 0$	Robust location estimation
Sign used	Uncentred: $Z_i = X_i / \ X_i\ $	Centred: $S(X_i; \mu) = (X_i - \mu) / X_i - \mu $
Weight	None ($W \equiv 1$)	Depth-based W_{PD} or W_{HSD}
Combination	Inner product $Z_i^\top Z_j$ (pairwise)	Weighted mean via IRLS
Output	Scalar $T_n \rightarrow$ p-value	Location estimator $\hat{q}_{nW}$
Dimension regime	$p \gg n$ explicit; martingale CLT	Fixed p ; $p^2/n \rightarrow 0$ as future work
Robustness	Signs collapse to unit sphere	Bounded IF (Proposition 1) when W bounded
Efficiency	ARE ≈ 2.54 vs Hotelling (t_3)	ARE > 1 vs unweighted median
Null distribution	$T_n / \sqrt{\text{var}(T_n)} \rightarrow N(0, 1)$	$\sqrt{n}(\hat{q}_{nW} - q_0) \rightarrow N(0, V_W)$
Key theorem	Lemma 3 (concentration on unit sphere)	Theorem 2 ($\text{eigvec}(\tilde{\Sigma}) = \text{eigvec}(\Sigma)$)
Shared root	Both from Möttönen & Oja (1995). JMVA SCM ($W \equiv 1$) is the kernel that JASA's T_n accumulates pairwise.	

Impact and Contribution

Impact and Contribution

What is the real-world impact of these two papers?

Impact of JASA (Wang, Peng & Li 2015)

1. **Extends nonparametric testing to $p \gg n$:** Before this paper, spatial sign tests were undefined when $p > n$. The paper shows they remain valid and powerful in high dimensions.
2. **Power gain under heavy tails grows with p :** A remarkable theoretical finding — the ARE of the new test vs CQ *increases* as p grows. For t_3 : $\text{ARE} \approx 2.54$ at large p vs 2.09 at $p = 10$ (fixed p).
3. **Application in genomics:** The paper applies the test to gene-set analysis with 15 diabetic patients and up to $p = 1607$ genes. The new test identifies 954 significant gene sets (37.87% of candidates) vs 520 for CQ (20.64%), discovering biologically relevant pathways that CQ misses.
4. **Dense alternatives:** Unlike sparse testing methods (Bonferroni, FDR), the new test works well when many components contribute small signals — the μ_2 (mixed alternative) results show this clearly.

Impact of JMVA (Majumdar & Chatterjee 2022)

1. **Resolves the low-efficiency problem of plain signs:** The plain SCM is asymptotically inadmissible (Magyar & Tyler 2014). Weighted signs fix this while keeping the bounded influence function.
2. **Unified framework:** The paper shows that $W \equiv 1$ (SCM) and $W = |X_i - \mu|$ (raw data) are both special cases. Any weight in between gives a robustness–efficiency trade-off the analyst can control.
3. **Applications beyond PCA:** The paper extends the framework to sufficient dimension reduction (SDR) and functional outlier detection (El Niño, octane data).
4. **Outlier detection:** Using robust PCA from $\tilde{\Sigma}$, the paper correctly identifies El Niño events (1982–83 and 1997–98) in sea-surface temperature data and all 6 alcohol-containing outliers in the octane spectroscopy dataset.

Impact of this replication study

1. **Connects the two papers:** Shows that JASA T_n and JMVA $\hat{q}_{nW}$ both gain power over CQ under heavy tails, and that the efficiency gains are comparable in the direct comparison (Module 3).
2. **Validates the open question:** Identifies the key open problem — whether a weighted version of the JASA T_n (replacing Z_i with $R_i = S(X_i; \mu) \cdot W_i$) would still admit a normal limit when $p \gg n$, given that $\|R_i\| = W_i \neq 1$ breaks the identity used in JASA’s proof.
3. **Practical guidance:** The contamination results show that W_{HSD} fails under extreme shifts ($\text{FSE} \approx 0.13$) while W_{PD} is more stable. This informs practitioners on which weight function to use.

Question for the Professor

Open Question at the Boundary of Both Papers

“In the JASA paper, T_n achieves a normal limit when $p \gg n$ via the martingale CLT. The proof relies on $\|Z_i\| = 1$ for all i , which enables the variance identity $T_n = (\|\sum Z_i\|^2 - n)/2$ and the concentration inequality in Lemma 3 (El Karoui 2009). In the JMVA paper, the weighted sign $R_i = S(X_i; \mu) \cdot W_i$ has norm $\|R_i\| = W_i \in (0, 1]$, not 1. If we replace Z_i with R_i in the JASA statistic:

$$T_n^W = \sum_i \sum_{j < i} R_i^\top R_j$$

would this still admit a tractable normal limit in the $p \gg n$ regime, or does the concentration inequality in Lemma 3 fundamentally require the signs to sit exactly on the unit sphere?”

This question shows you have read both papers carefully and understood the technical mechanism that makes each proof work. The variance identity and Lemma 3 are used explicitly in JASA’s proof — breaking $\|Z_i\| = 1$ requires rebuilding both.

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